

DOES A QUIETER CITY MEAN FEWER COMPLAINTS?

The Sounds of New York City During COVID-19 Lockdown

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Effects of COVID-19 outbreak on the sound environment

COVID-19

COVID-19 lockdowns have silenced urban noise. Now it's coming back

May 27, 2020

Article
Quieted City Sounds during the COVID-19 Pandemic in Montreal

Assessing the changing urban sound environment during the COVID-19 lockdown period using short-term acoustic measurements

OPEN Indoor and outdoor noise changes due to the COVID-19 lockdown and the expected

COVID-19 pandemic causes a seismic noise quiet period in 2020

How the Pandemic Changed the Urban Soundscape

A map of recordings from around the world during the pandemic, showing a period like without noise pollution

Coronavirus Turns Urban Life's Roar Deafening on World's Seismographs

Scientists are getting a clearer picture of what's going on with the planet as lockdowns keep many of us at home.

Check for updates

OPEN Impact of the COVID-19 pandemic on levels of deep-ocean acoustic noise

The New York Times

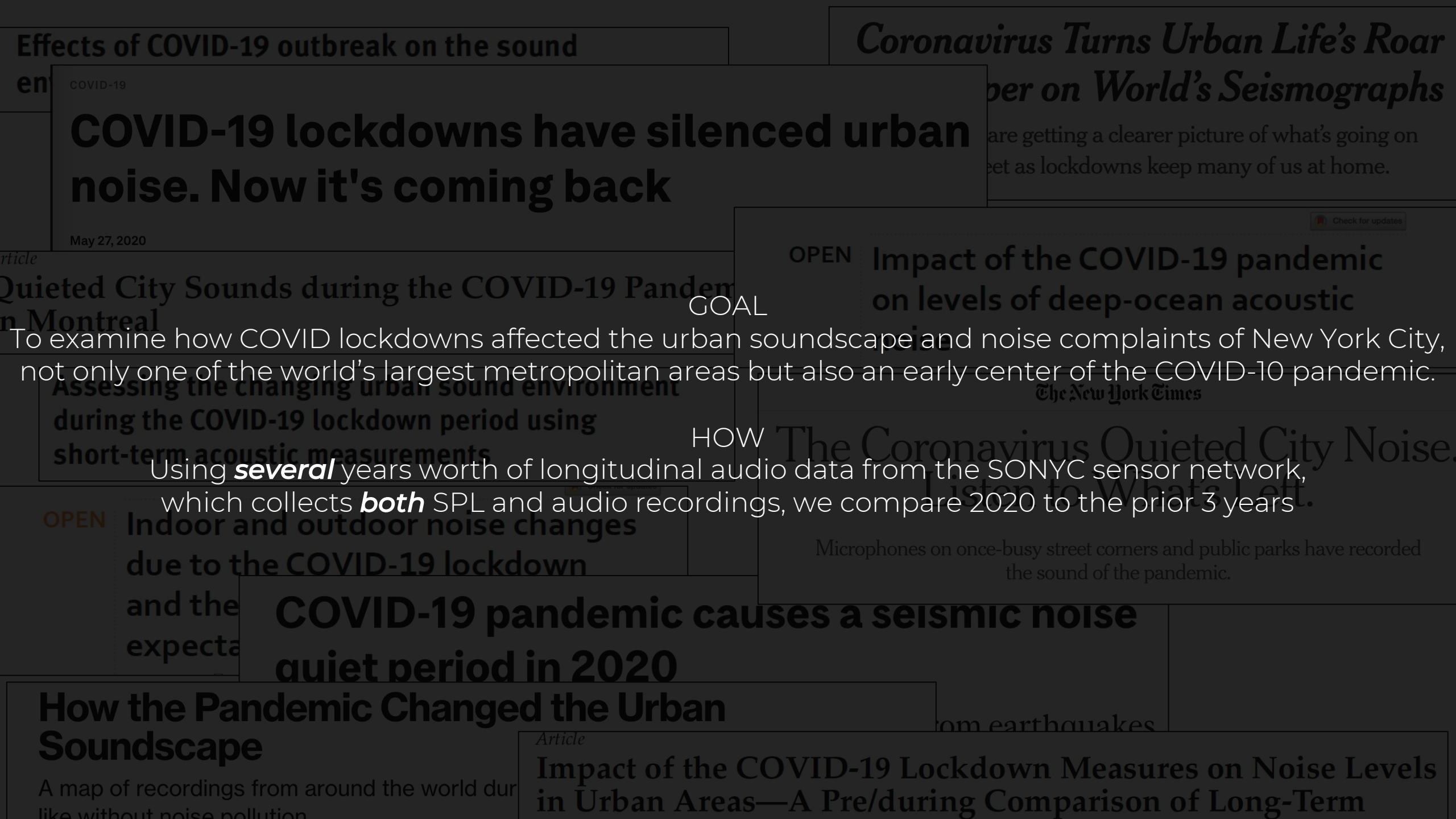
The Coronavirus Quieted City Noise. Listen to What's Left.

Microphones on once-busy street corners and public parks have recorded the sound of the pandemic.

Article

Impact of the COVID-19 Lockdown Measures on Noise Levels in Urban Areas—A Pre/during Comparison of Long-Term

from earthquakes



Effects of COVID-19 outbreak on the sound

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Quieted City Sounds during the COVID-19 Pandemic in Montreal

To examine how COVID lockdowns affected the urban soundscape and noise complaints of New York City, not only one of the world's largest metropolitan areas but also an early center of the COVID-10 pandemic.

Assessing the changing urban sound environment during the COVID-19 lockdown period using short-term acoustic measurements

OPEN

Impact of the COVID-19 pandemic on levels of deep-ocean acoustic

GOAL

HOW

Using **several** years worth of longitudinal audio data from the SONYC sensor network, which collects **both** SPL and audio recordings, we compare 2020 to the prior 3 years

The New York Times

The Coronavirus Quieted City Noise. Listen to What's Left.

OPEN

Indoor and outdoor noise changes due to the COVID-19 lockdown and the expected impact on the urban soundscape

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THE SONYC SENSOR NETWORK

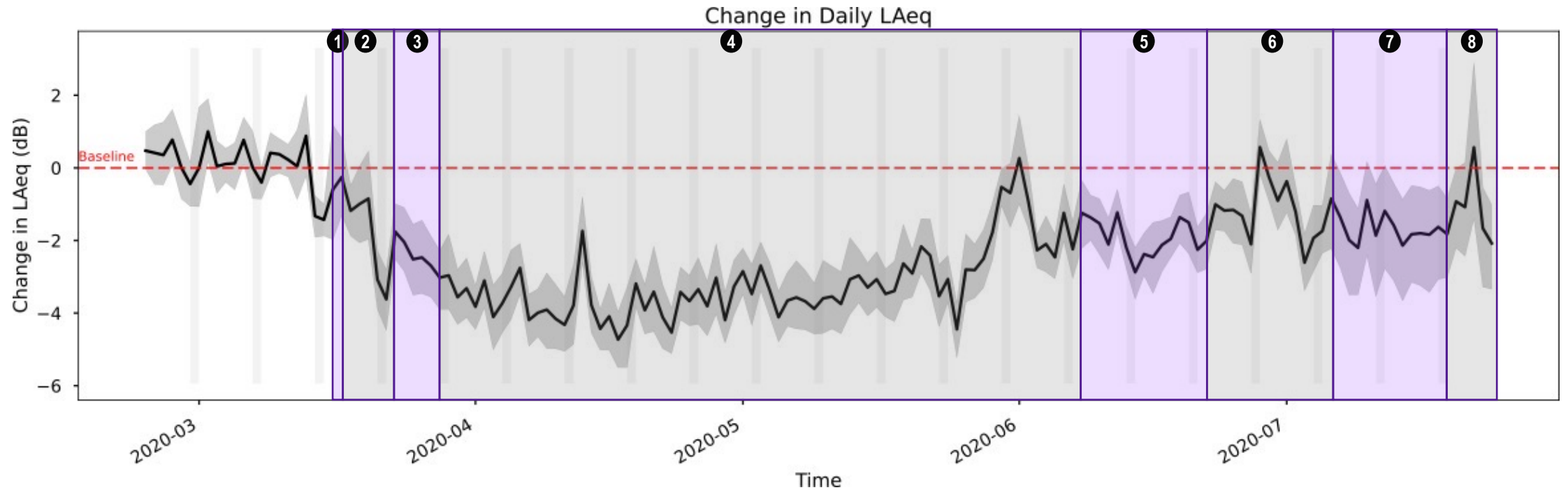
- Active in 2016–2022
- 75 low-cost acoustic sensors
- Deployed in 3 NYC boroughs
- Sensors typically mounted 15–25 feet above ground
- Aim was to facilitate continuous, real-time, accurate, and source-specific monitoring of noise
- Cumulatively collected:
 - 150 years of calibrated SPL data
 - 51 years of raw audio snippets



DATASET USED FOR ANALYSIS

- Data from 11 sensors active from early 2017 through July 24, 2020
- Analysis focused on Feb. 24, 2020 to July 24, 2020, encompassing lead up to NYC lockdown and Phase 4 reopening
- Data from 2017–2019 was used as a baseline to which to compare the target period
- Sensor locations had. variety of characteristics as shown on the table on the right.

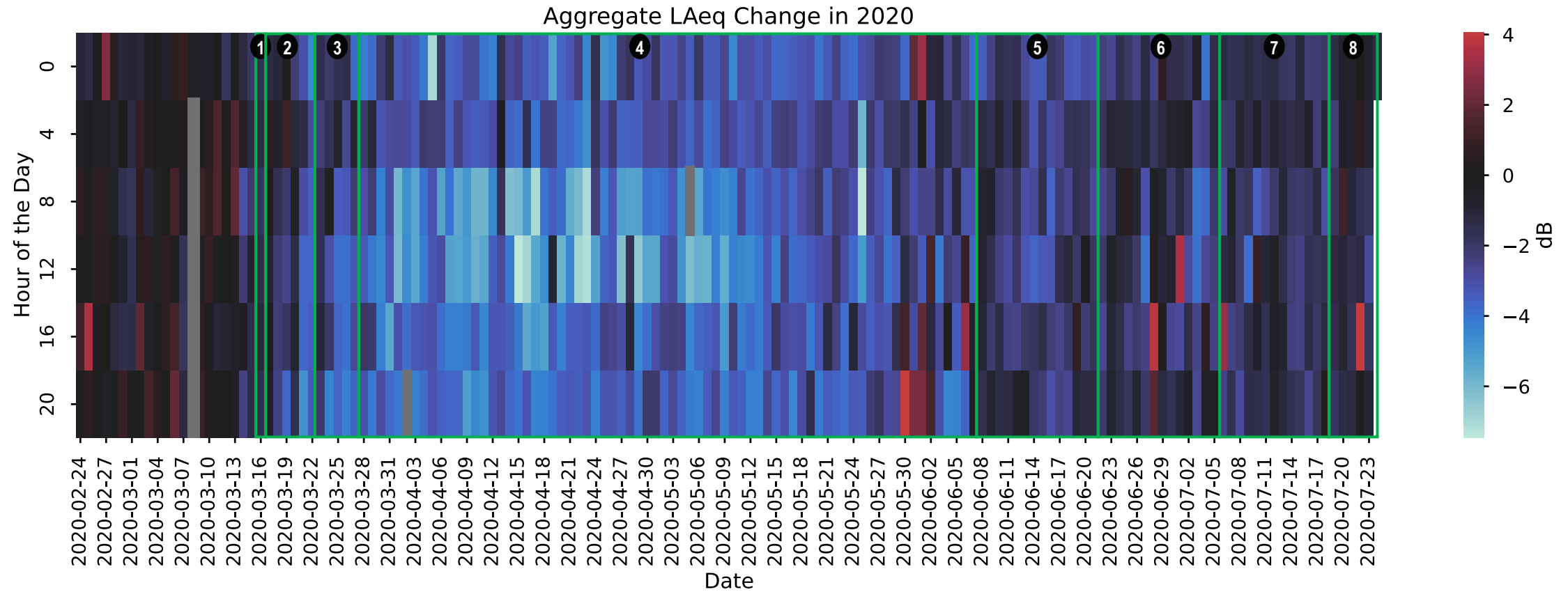
AN UNPRECEDENTED DROP IN NOISE



1. Schools closed
2. Bars/restaurants close
3. New York State on pause
4. Construction halt

5. Phase 1 reopening (construction, manufacturing)
6. Phase 2 reopening (outdoor, dining, retail)
7. Phase 3 reopening (indoor dining @ reduced capacity)
8. Phase 4 reopening (low-risk outdoor activities @ reduced capacity)

THE PATTERNS OF CHANGE IN NOISE LEVEL

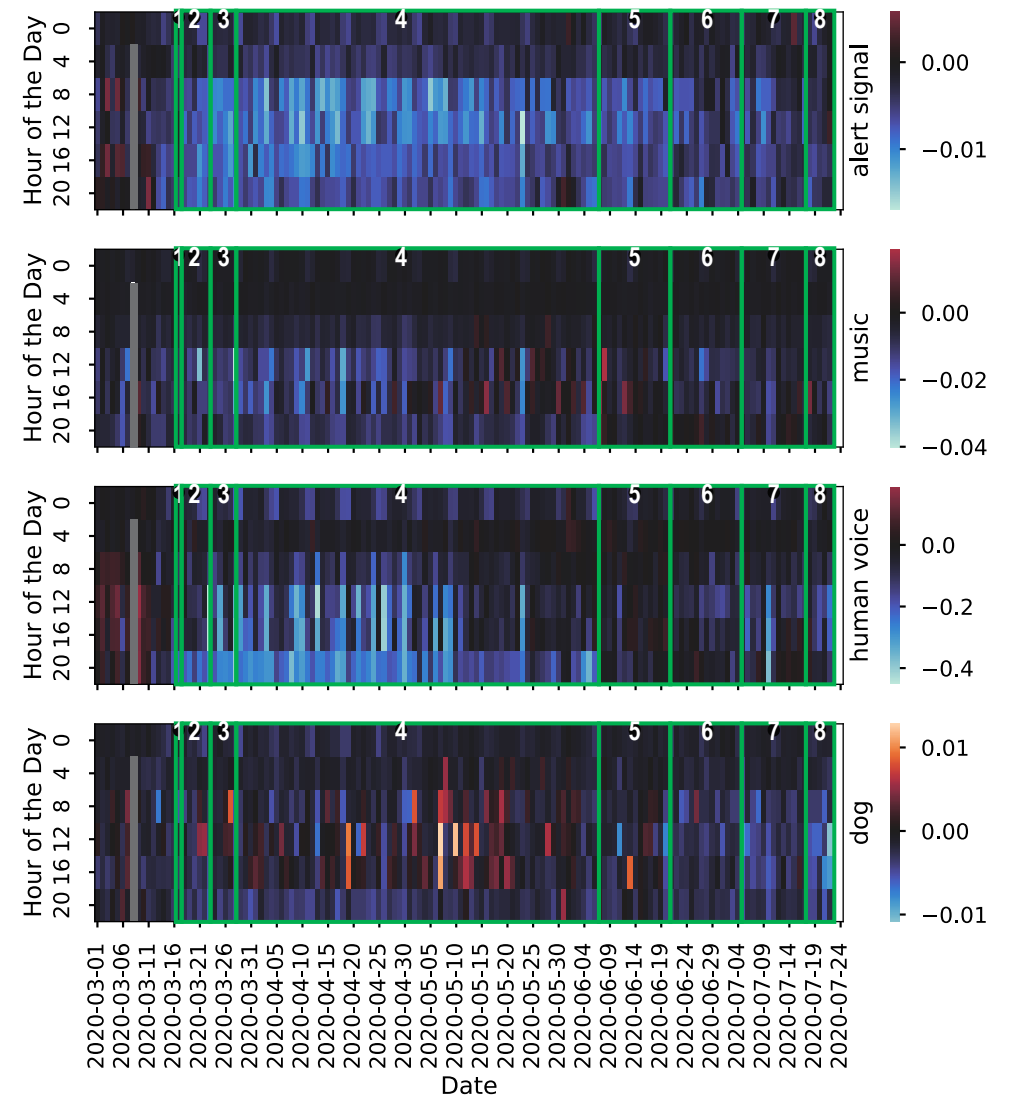
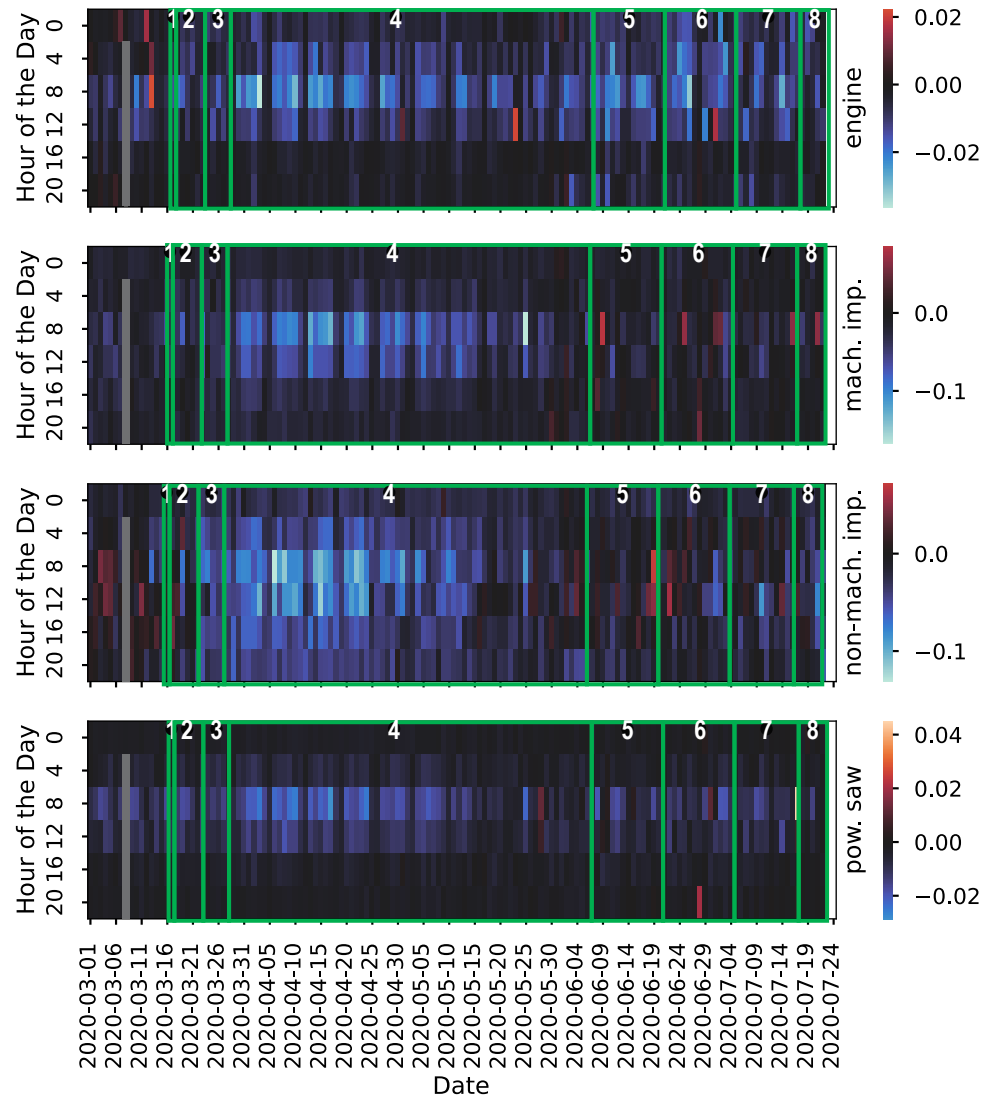


THE PATTERNS OF CHANGE IN NOISE *SOURCES*

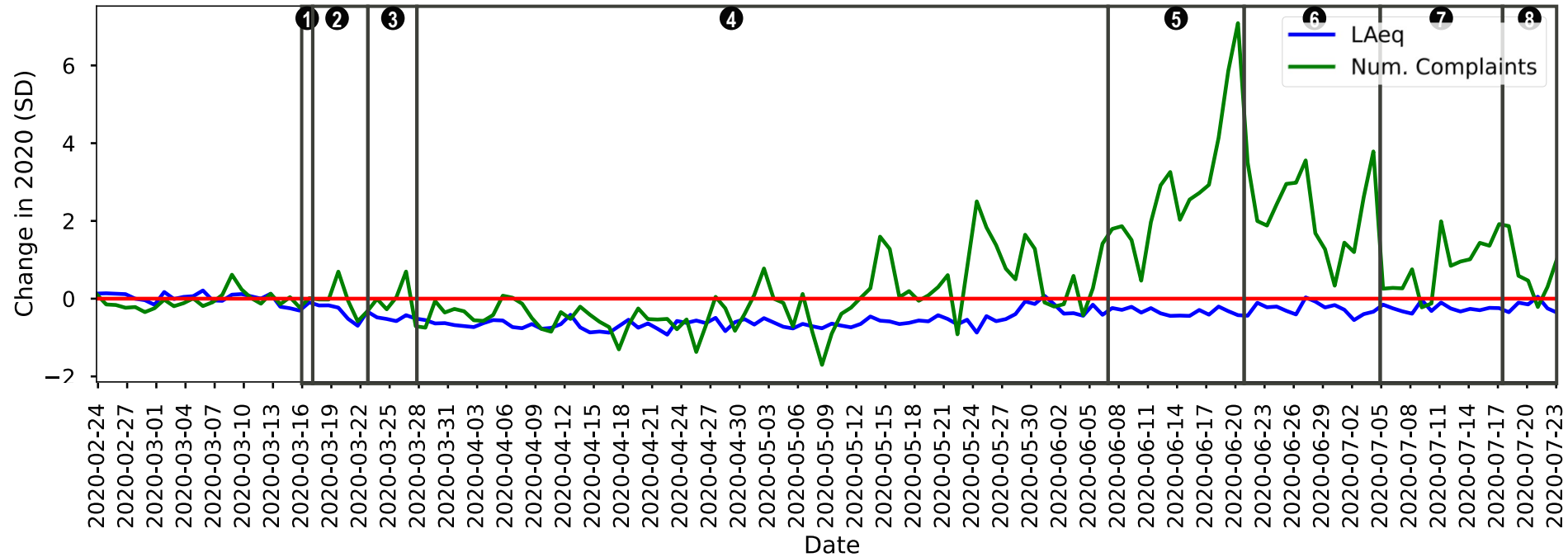
Model:

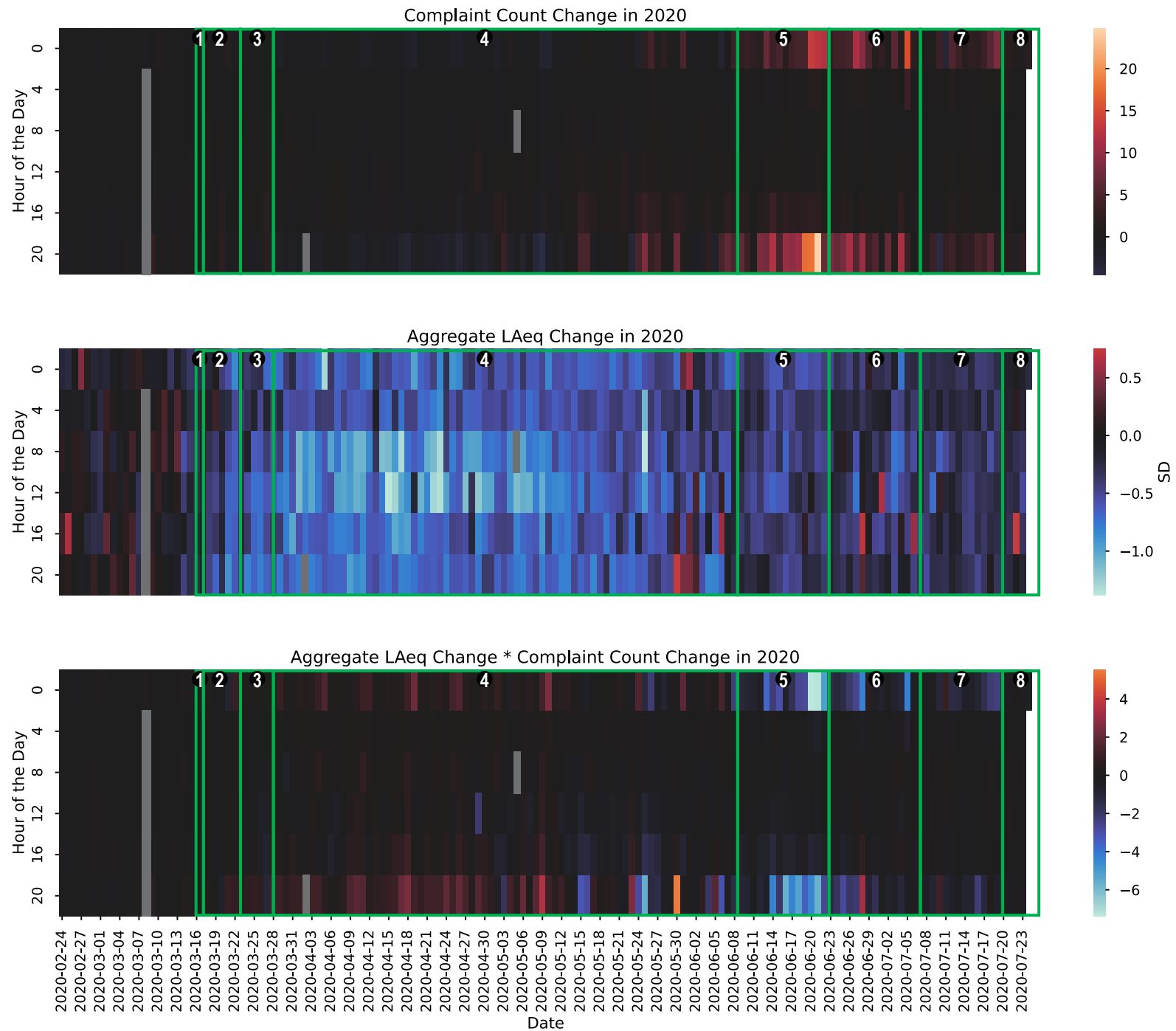
- Multi-label multi-layer perception w/ 1 hidden layer with 128 ReLU units
- Input: 512-dim OpenL3 features representing for each 1 s frame, 0.5 s hop size
- AutoPool temporal pooling for training on 10 s audio clips
- Output: 8 class likelihoods:
engine, machinery impact, non-machinery impact, powered saw, alert signal, music, human voice, dog
- Trained on SONYC Urban Sound Tagging (UST) Dataset v2.3

THE PATTERNS OF CHANGE IN NOISE SOURCES



A QUIET CITY DOES NOT MEAN FEWER COMPLAINTS





SUMMARY

- Using several years of data from the SONYC acoustic sensor network, we analyzed the noise level, noise class, and complaint patterns of the COVID-related lockdown and recovery in New York City
- Some results confirmed those from other cities, but our extensive historical data was able to give us insights into the patterns and relationships of these three forms of data and confirm that noise and complaints are not always correlated, which is often difficult to demonstrate with data.
- Models predict sound class presence, but that does not quantify impact on noise. We need robust models to predict source-specific loudness.
- Fixed vocabulary models focusing on urban sound classes, but the unexpected is sometimes most interesting. What patterns may we have missed in these unusual circumstances?